Lacunary sequences whose reciprocal sums represent all rational numbers in an interval

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Abstract. This is the blueprint of a simplified version of my paper with Vjeko [2] solving Erdős Problem #355 [3]. This blueprint was written with the help of Gemini3 in order to get Aristotle from Harmonic to formalize it into Lean [1].

Initial draft generated by Gemini3 in this chat:
https://gemini.google.com/share/0f455718e685

Reworked by Vjeko and Gemini3 in this chat:
https://gemini.google.com/share/5ece3d9fcef9

1. Formulation

Definition 1. *For a sequence of positive real numbers $(x_i)_{i=1}^{\infty}$ define $P((x_i)_{i=1}^{\infty})$ as the set*

$$P((x_i)_{i=1}^{\infty}) := \{\sum_{i\in T} x_i : T \text{ is a finite subset of } \mathbb{N}\}.$$

Definition 2. *A sequence of positive integers $(a_i)_{i=1}^{\infty}$ is said to be lacunary if there exists a real number $\lambda > 1$ such that $a_{i+1}/a_i \geq \lambda$ for every $i \geq 1$.*

Theorem 1. *There exists a lacunary sequence of positive integers $(a_i)_{i=1}^{\infty}$ and real numbers α, β such that $0 \leq \alpha < \beta$ and the set $P((1/a_i)_{i=1}^{\infty})$ contains all rational numbers in the interval (α, β).*

2. Goal

We prove that there exists a strictly increasing sequence of positive integers $(a_i)_{i=1}^{\infty}$ such that $a_{i+1}/a_i \geq 1.01$ for every index $i \geq 1$ and the set $P((1/a_i)_{i=1}^{\infty})$ contains all rational numbers in the interval $(0, 2)$. This will prove Theorem 1 by taking $\lambda = 1.01$, $\alpha = 0$, $\beta = 2$.

3. Definitions and Tools

3.1. **Finite Sequences (Blocks).** We define a *block* as a finite sequence of integers $B = (b_0, b_1, \ldots, b_m)$. We denote the length of the block by $|B| = m + 1$. The last element is denoted $\text{last}(B) = b_m$.

Lemma 2 (Block Existence)**.** *For every integer $k \geq 2$, there exists a finite sequence of integers $D_k = (d_0, d_1, \ldots, d_{m_k})$ satisfying the following properties:*

(1) ***Start/End:*** *$d_0 = 1$ and d_{m_k} is divisible by k.*
(2) ***Divisibility:*** *For all $0 \leq j < m_k$, d_j divides d_{m_k}.*

(3) ***Growth:*** *For all* $0 \le j < m_k$, $1.01 \le \frac{d_{j+1}}{d_j} \le 2$.

Proof. Let $a = \lfloor \log_2 k \rfloor$. We construct D_k based on the value of k:

- **Case 1 ($k = 2^a$):** Set $m_k = a$. Define $d_j = 2^j$.
- **Case 2 ($1 < k/2^a < 1.1$):** Set $N = 3k \cdot 2^{a+1}$. Define the sequence by enumerating powers of 2 up to 2^{a+1}, then jumping to N.
- **Case 3 ($k/2^a \ge 1.1$):** Set $N = k \cdot 2^a$. Define the sequence by enumerating powers of 2 up to 2^a, then jumping to N.

□

3.2. Rational Representation in Finite Sets.

Lemma 3 (Finite Reachability)**.** *Let* $A = (x_1, \dots, x_N)$ *be a sequence of integers such that* $x_1 | x_N$, $x_2 | x_N$, $\dots$, $x_{N-1} | x_N$. *If* $1 < \frac{x_{i+1}}{x_i} \le 2$ *for all* $1 \le i < N$, *then*

$$P((1/x_i)_{i=1}^N) = \left\{ \frac{u}{x_N} \;\middle|\; u \in \mathbb{Z}_{\ge 0}, \quad 0 \le \frac{u}{x_N} \le \sum_{i=1}^{N} \frac{1}{x_i} \right\}.$$

Proof. Consider the sequence $b_i = x_N / x_{N+1-i}$. We have $b_1 = 1$ and $b_{i+1} \le 2b_i$. By Brown's Criterion (or simple induction on reachable integer sums), the set of subset sums of $\{b_i\}$ is the complete interval of integers $\{0, \dots, \sum b_i\}$. Dividing by x_N yields the result. □

4. The Construction

We define the infinite sequence $(a_i)_{i=1}^{\infty}$ by concatenating scaled versions of the blocks D_k from Lemma 2.

Definition 3 (The Sequence)**.** *Let* $D_k = (d_{k,0}, \dots, d_{k,m_k})$ *be the block for integer* k *derived in Lemma 2. We define the sequence of blocks* $(B_k)_{k=1}^{\infty}$ *recursively:*

- $B_1 = (1)$.
- *For* $k \ge 2$, *let* $L_{k-1} = last(B_{k-1})$. *We define* B_k *as the sequence* D_k *scaled by* L_{k-1}*:*

  $$B_k = (L_{k-1} \cdot d_{k,1}, \quad L_{k-1} \cdot d_{k,2}, \quad \dots, \quad L_{k-1} \cdot d_{k,m_k}).$$

  (Note: We omit $d_{k,0} = 1$ *to avoid duplicating the join point, as* $L_{k-1} \cdot d_{k,0} = L_{k-1}$*).*

The sequence (a_i) *is the concatenation* $B_1 \frown B_2 \frown B_3 \dots$

5. Verification

5.1. Lacunarity.

We check the ratio a_{i+1}/a_i for all i.

- **Inside a block B_k:** The terms are $L_{k-1} d_{k,j}$. The ratio is $d_{k,j+1}/d_{k,j}$, which is in $[1.01, 2]$ by Lemma 2.

- **At the boundary of B_{k-1} and B_k:** Let $u = \text{last}(B_{k-1}) = L_{k-1}$. The first term of B_k is $v = L_{k-1}d_{k,1}$. The ratio is $d_{k,1}/d_{k,0} = d_{k,1}$, which is in $[1.01, 2]$ by Lemma 2.

Thus, (a_i) is lacunary with $\lambda = 1.01$.

5.2. **Divisibility Chain.** Let $end(k)$ be the index of the last term of block B_k in the full sequence. By definition, $a_{end(k)} = \text{last}(B_k) = \text{last}(B_{k-1}) \cdot \text{last}(D_k)$. Since $\text{last}(D_k)$ is divisible by k (Lemma 2), we have:

$$k \mid a_{end(k)}.$$

Furthermore, by construction, $a_i \mid a_{i+1}$ is not necessarily true, but $a_i \mid a_{end(k)}$ holds for all $i \leq end(k)$ because: 1. Within block k, $d_{k,j} \mid d_{k,m_k}$. 2. Between blocks, the scaling factor ensures transitivity.

5.3. **Covering Rationals.** Let $q \in \mathbb{Q} \cap (0, 2)$. Write $q = u/v$ with $u, v \in \mathbb{Z}^+$.

(1) Choose k large enough such that $k \geq v$ and $\sum_{i=1}^{end(k)} \frac{1}{a_i} > q$. (The sum diverges because $a_i \leq 2^{i-1}$, so the harmonic series limit is ≥ 2).
(2) Consider the prefix of the sequence up to index $N = end(k)$.
(3) By the divisibility property, $v \mid k \mid a_N$, so q can be written as U/a_N for some integer U.
(4) By Lemma 3, the set of reciprocal sums of the prefix $\{a_1, \ldots, a_N\}$ contains all fractions of the form X/a_N up to the total sum.
(5) Since $q < \text{Sum}$ and $q = U/a_N$, q is represented.

References

[1] ErdősProblem#355.lean. https://github.com/Woett/Lean-files/tree/main.
[2] Wouter van Doorn and Vjekoslav Kovač, *Lacunary sequences whose reciprocal sums represent all rational numbers in an interval.* arXiv:2509.24971 (2025).
[3] T. F. Bloom, Erdős Problem #355, https://www.erdosproblems.com, accessed 2026-01-21.